# IGCSE 0607 Practice Questions — E5.7 — Combined Questions & Markscheme

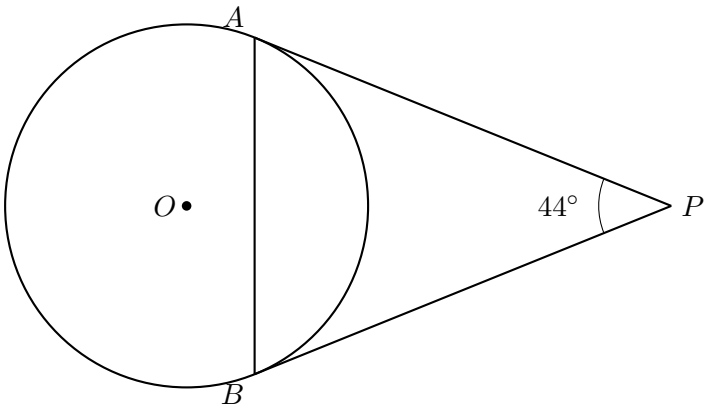
## Circle theorems II

**Total Questions:** 24 — **Total Marks:** 156 Marks

## Table of Contents & Quick Navigation


- **E5.7** Circle theorems II
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*

**1** The diagram shows a circle with centre $O$.
$P$ is an external point.
The lines $PA$ and $PB$ are tangents to the circle at the points $A$ and $B$.
Angle $APB = 44°$.




NOT TO SCALE

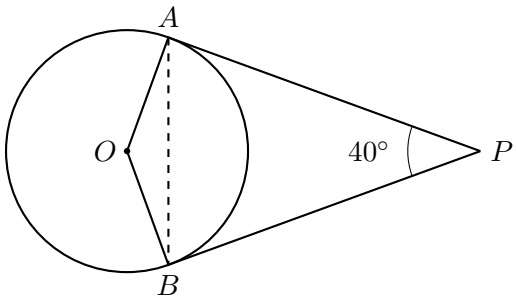
**(a)** Calculate angle $PAB$.

Angle $PAB =$ ..... ° **[2]**

**(b)** Calculate angle $AOB$.

Angle $AOB =$ ..... ° **[2]**

**2** $P$ is a point outside a circle, centre $O$.
$PA$ and $PB$ are tangents to the circle at $A$ and $B$.
Angle $APB = 40°$.




**(a)** Write down the geometrical reason why triangle $PAB$ is isosceles.

..... **[1]**

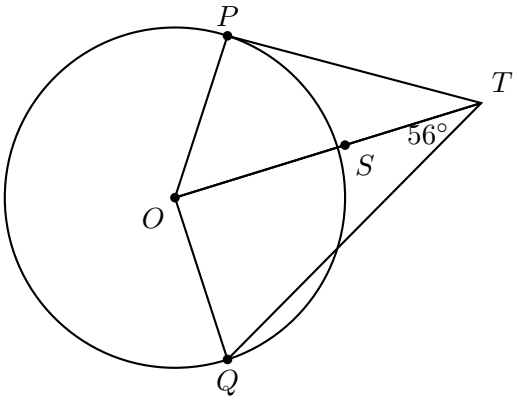
**(b)** Calculate angle $PAB$.

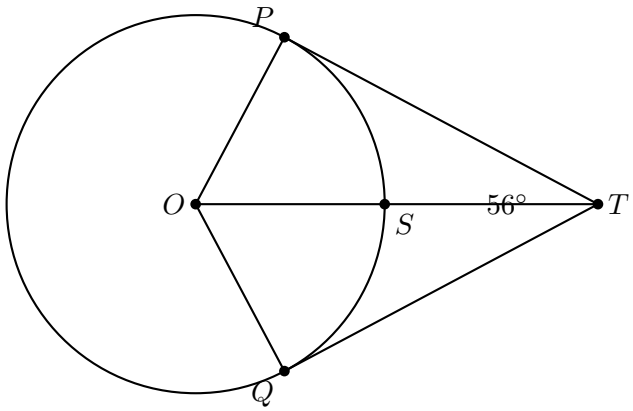
Angle $PAB =$ ..... ° **[1]**

**(c)** Calculate angle $AOB$.

Angle $AOB =$ ..... ° **[2]**

**3** $P$ and $Q$ are points on a circle, centre $O$.
$TP$ and $TQ$ are tangents to the circle from an external point $T$.
The line $TO$ intersects the circle at $S$.
Angle $PTQ = 56°$.




**(a)** Calculate angle $POT$.

Angle $POT =$ ..... ° **[2]**

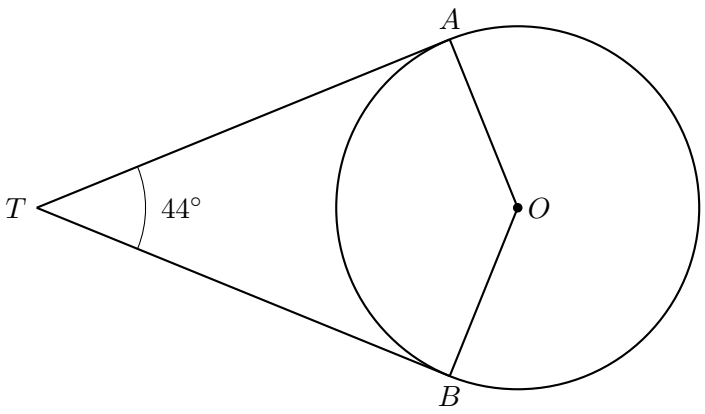
**(b)** Explain why triangle $TPQ$ is isosceles.

..... **[1]**

**(c)** Calculate angle $TPQ$.

Angle $TPQ =$ ..... ° **[1]**

**4** The diagram shows a circle, centre $O$.
$TA$ and $TB$ are tangents to the circle at $A$ and $B$.
Angle $ATB = 44°$.




**(a)** Work out angle $AOB$.

Angle $AOB =$ ..... ° **[2]**

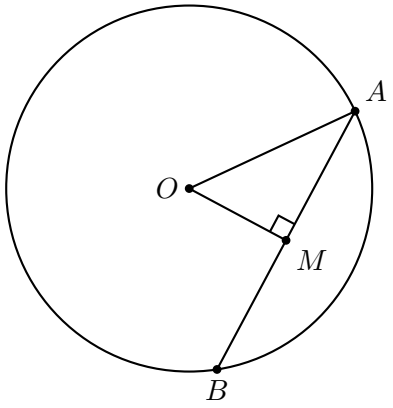
**(b)** Work out angle $OAB$.

Angle $OAB =$ ..... ° **[2]**

**(c)** Write down the geometrical reason why the length of $TA$ is equal to the length of $TB$.

..... **[1]**

**5** The diagram shows a circle, centre $O$, with radius 10 cm.
$AB$ is a chord of length 16 cm and $M$ is the midpoint of $AB$.



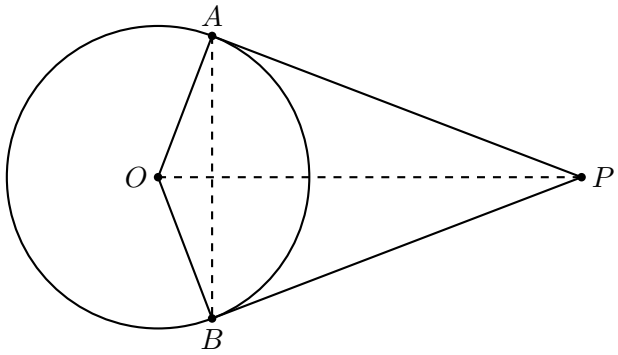

**(a)** State the size of angle $OMA$, giving a geometrical reason for your answer.

$\angle OMA =$ ..... ° **[2]**

**(b)** Calculate the distance $OM$ of the chord $AB$ from the centre $O$.

$OM =$ ..... $cm$ **[3]**

**6** $P$ is an external point.
$PA$ and $PB$ are tangents to a circle, centre $O$.
The line $OP$ intersects the chord $AB$ at right angles.
$OA = 7$ cm and $PA = 24$ cm.




**(a)** Write down the length of $PB$.

..... cm **[1]**

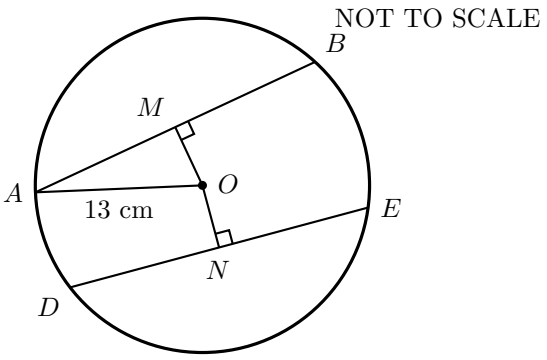
**(b)** Calculate the length of $OP$.

..... cm **[2]**

**(c)** Calculate the shortest distance from $O$ to the chord $AB$.

..... cm **[3]**

**7** The diagram shows a circle, centre $O$, of radius 13 cm. Chords $AB$ and $DE$ each have length 24 cm. The line $OM$ is perpendicular to $AB$ at $M$, and the line $ON$ is perpendicular to $DE$ at $N$.




**(a)** Write down the length of $AM$ and give a geometrical reason for your answer.

$AM =$ ..... , Reason: ..... **[2]**

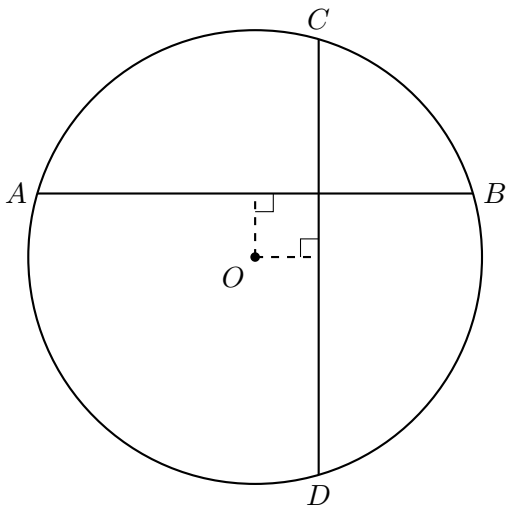
**(b)** Calculate the distance $OM$.

$OM =$ ..... cm **[2]**

**(c)** Write down the length of $ON$ and give a geometrical reason for your answer.

$ON =$ ..... , Reason: ..... **[2]**

**8** The diagram shows a circle, centre $O$, with radius 25 cm.
$AB$ and $CD$ are two chords of the circle such that $AB = CD$.
The perpendicular distance from $O$ to $AB$ is $(x + 3)$ cm.
The perpendicular distance from $O$ to $CD$ is $(3x - 5)$ cm.



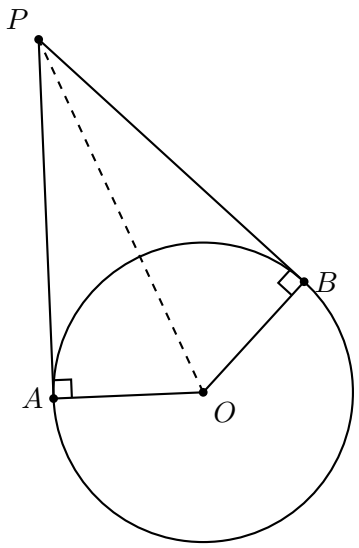

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Calculate the length of the chord $AB$.

$AB =$ ..... cm **[3]**

**9** The diagram shows a circle, centre $O$, with radius 5 cm.
$PA$ and $PB$ are tangents to the circle at $A$ and $B$ respectively.
The length of $PA$ is 12 cm.




**(a)** Write down the length of $PB$, giving a geometrical reason for your answer.

$PB =$ ..... $cm$ **[2]**

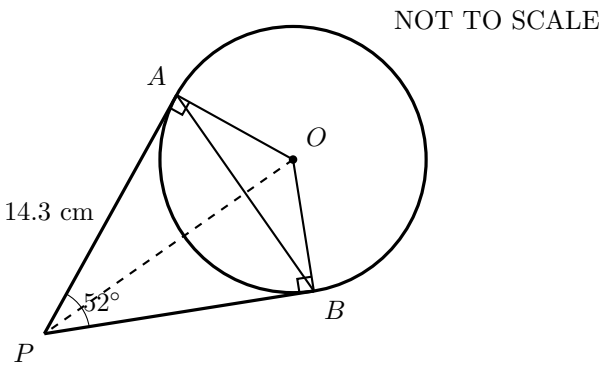
**(b)** Calculate the distance $OP$.

$OP =$ ..... $cm$ **[2]**

**(c)** Calculate the area of the quadrilateral $OAPB$.

Area = ..... $cm^2$ **[2]**

**10** The diagram shows a circle, centre $O$, and an external point $P$.
The straight lines $PA$ and $PB$ are tangents to the circle at $A$ and $B$ respectively.
Angle $APB = 52°$ and $PA = 14.3$ cm.




**(a)** Explain why triangle $OAP$ is congruent to triangle $OBP$.

.....

.....

..... **[3]**

**(b)** Calculate the size of

**(i)** angle $AOB$,

Angle $AOB =$ ..... ° **[2]**

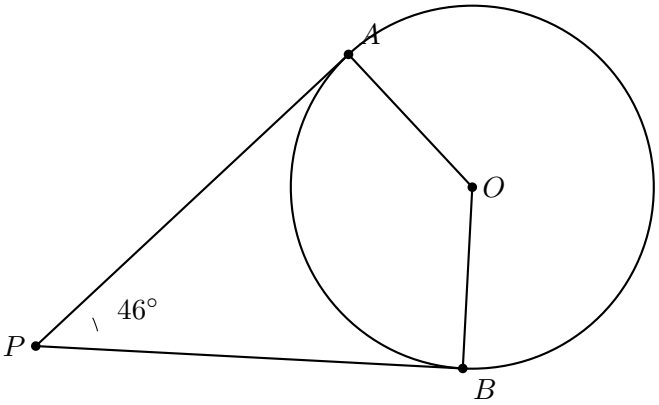
**(ii)** angle $OAB$.

Angle $OAB =$ ..... ° **[2]**

**(c)** Write down the length of $PB$.

$PB =$ ..... cm **[1]**

**11** $P$ is an external point.
$PA$ and $PB$ are tangents to the circle, centre $O$, at the points $A$ and $B$.
Angle $APB = 46°$.



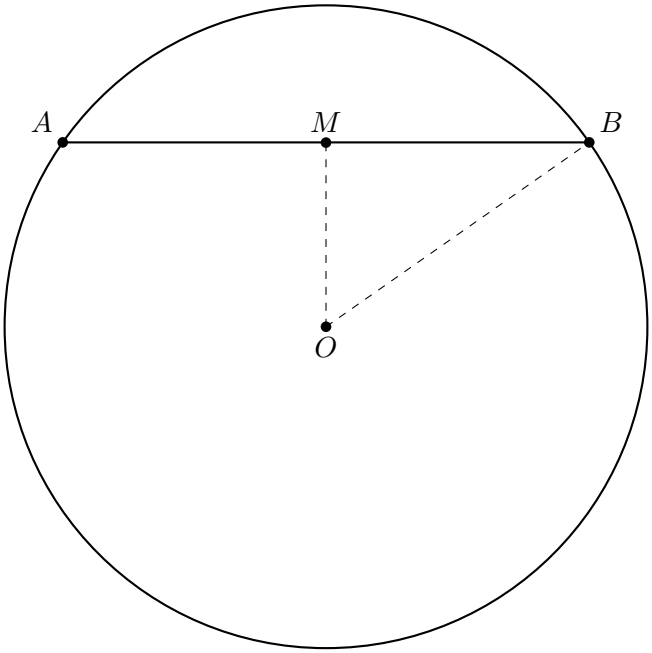

**(a)** Calculate angle $AOB$.

Angle $AOB$ = ..... ° **[2]**

**(b)** Calculate angle $OAB$.

Angle $OAB$ = ..... ° **[2]**

**12** A circle, centre $O$, has a radius of 8.5 cm.
A chord $AB$ has length 14 cm.
$M$ is the midpoint of $AB$.



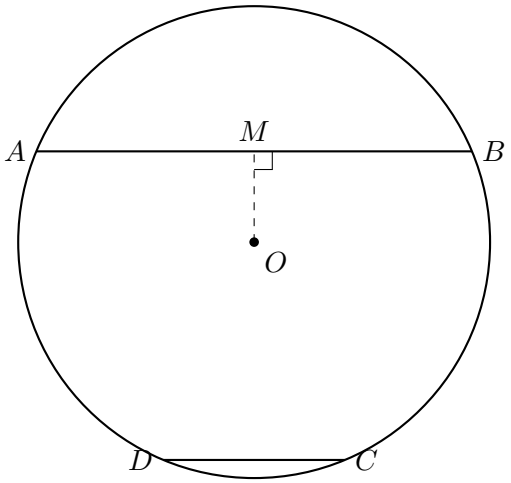

**(a)** Calculate the distance $OM$.

$OM =$ ..... cm **[3]**

**(b)** Another chord $CD$ is drawn such that it is parallel to $AB$ and is at a distance of 3 cm from $O$.
Calculate the length of the chord $CD$.

$CD =$ ..... cm **[2]**

**13** The diagram shows a circle, centre $O$, with radius 13 cm.
$AB$ is a chord of the circle of length 24 cm.
The point $M$ is the midpoint of the chord $AB$.
Another chord $CD$ is drawn inside the circle.




NOT TO SCALE

**(a)** Calculate the perpendicular distance, $OM$, from the centre of the circle to the chord $AB$.

$OM =$ ..... cm **[3]**

**(b)** The perpendicular distance from the centre $O$ to the chord $CD$ is also 5 cm.
Write down the length of the chord $CD$.

$CD =$ ..... cm **[1]**

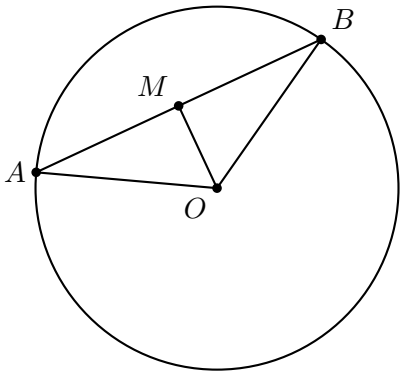
**(c)** Calculate the area of triangle $OAB$.

Area =

.....

$cm^2$ **[2]**

**14** $A$ and $B$ are points on a circle, centre $O$ and radius 13 cm.
The length of the chord $AB$ is 24 cm.
The point $M$ is the midpoint of $AB$.




**(a)** State the geometrical reason why angle $OMA = 90°$.

..... **[1]**

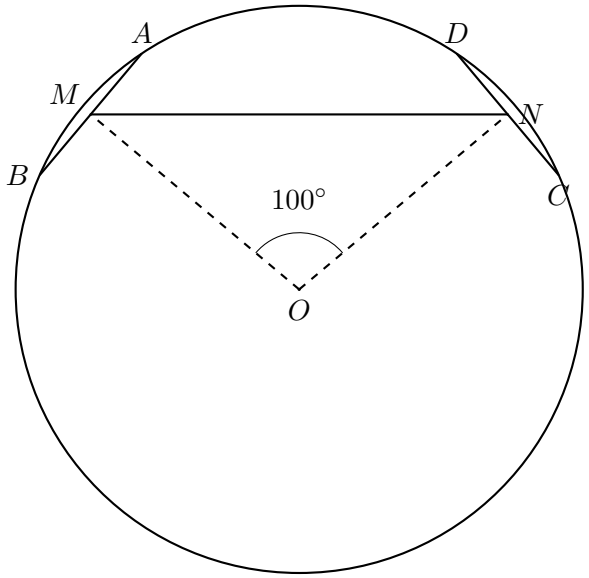
**(b)** Calculate the length of $OM$.

$OM =$ ..... cm **[3]**

**(c)** Calculate angle $AOB$.

Angle $AOB =$ ..... ° **[3]**

**15** $A$, $B$, $C$ and $D$ are points on a circle, centre $O$ and radius 25 cm.
The chords $AB$ and $CD$ each have a length of 14 cm.
$M$ is the midpoint of $AB$ and $N$ is the midpoint of $CD$.
Angle $MON = 100°$.




**(a)** Show that $OM = 24$ cm.

**[2]**

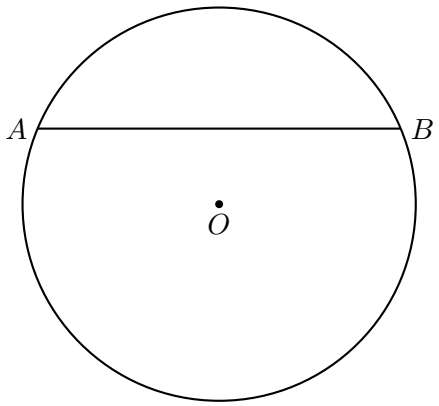
**(b)** Calculate the length of $MN$.

..... cm **[3]**

**(c)** Calculate the area of triangle $MON$.

..... $cm^2$ **[2]**

**16** The diagram shows a circle, centre $O$, with radius 13 cm.
$AB$ is a chord of the circle of length 24 cm.




**(a)** Calculate the shortest distance from $O$ to the chord $AB$.

..... *cm* **[3]**

**(b)** $CD$ is another chord of the circle.
$CD$ is parallel to $AB$ and has length 24 cm.
Write down the shortest distance from $O$ to $CD$.

..... *cm* **[1]**

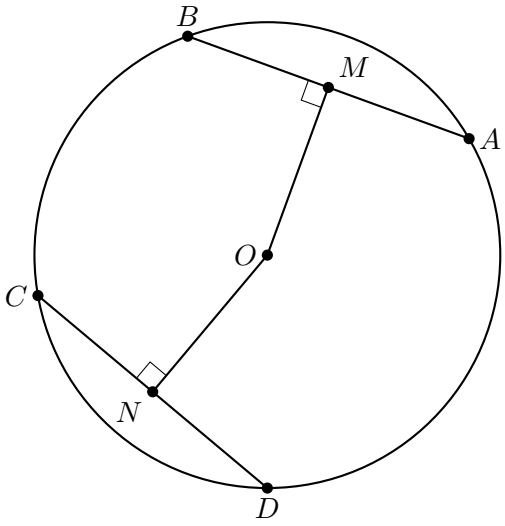
**(c)** $EF$ is a third chord of the circle.
The shortest distance from $O$ to $EF$ is 12 cm.
Calculate the length of the chord $EF$.

..... *cm* **[3]**

**(d)** State the geometrical property of a circle used to find the length of half the chord in part (a).

..... **[1]**

**17** $AB$ and $CD$ are two chords of a circle, centre $O$.
The chords $AB$ and $CD$ are equal in length.
The line $OM$ is the perpendicular bisector of $AB$, and the line $ON$ is the perpendicular bisector of $CD$.
$OM = (5x - 7)$ cm and $ON = (2x + 2)$ cm.
The radius of the circle is 17 cm.



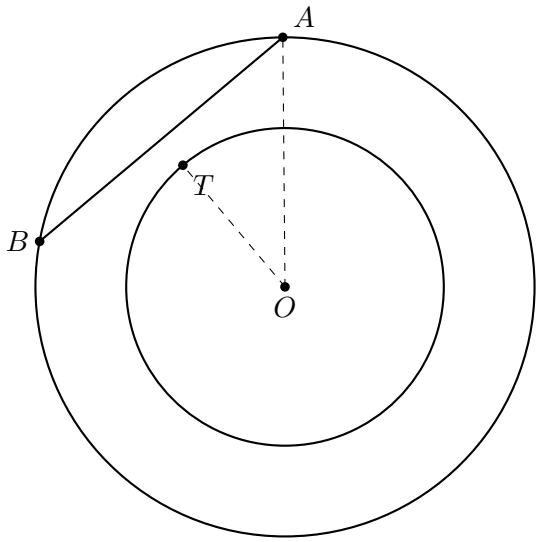

**(a)** Find the value of $x$.

$x =$ ..... **[3]**

**(b)** Calculate the length of the chord $AB$.

$AB =$ ..... cm **[4]**

**18** Two circles, $C_1$ and $C_2$, have a common centre $O$.
The radius of $C_1$ is $r$ cm and the radius of $C_2$ is $R$ cm, where $R > r$.
A chord $AB$ of the larger circle $C_2$ is tangent to the smaller circle $C_1$ at the point $T$.
The length of the chord $AB$ is 24 cm and the area of the annular region between the two circles is $k\pi$ cm$^2$.




**(a)** Show that $R^2 - r^2 = 144$.

**[3]**

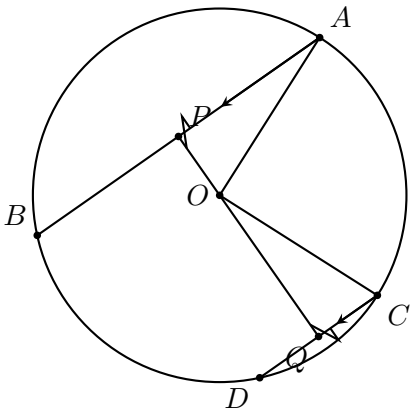
**(b)** Find the value of $k$.

$k =$ ..... **[2]**

**(c)** A second chord $CD$ of the larger circle $C_2$ has length 24 cm.
State, with a reason, the distance from $O$ to the chord $CD$.

..... **[2]**

**19** The diagram shows a circle, centre $O$, with two parallel chords $AB$ and $CD$ on opposite sides of the centre.
$AB = 24$ cm, $CD = 10$ cm, and the perpendicular distance between the two chords is 17 cm.
$P$ and $Q$ are the midpoints of $AB$ and $CD$ respectively.



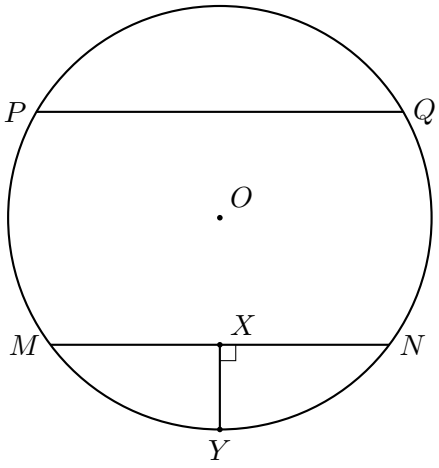

**(a)** Explain why $P$, $O$ and $Q$ lie on a straight line.

..... **[2]**

**(b)** Let $OP = x$ cm and the radius of the circle be $r$ cm.
Calculate the radius, $r$, of the circle.

$r =$ ..... $cm$ **[5]**

**20** A circle has centre $O$ and radius 10 cm.
$MN$ is a chord of length 16 cm.
The perpendicular bisector of $MN$ intersects $MN$ at $X$ and the minor arc $MN$ at $Y$.
$PQ$ is a second chord which is parallel to $MN$.
The perpendicular distance between $MN$ and $PQ$ is 11 cm.




**(a)** Explain why the line containing $XY$ passes through the centre $O$.

..... **[1]**

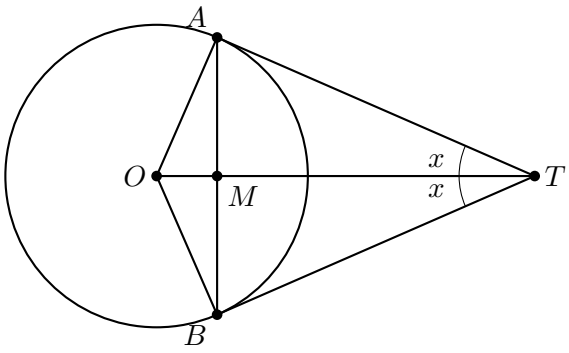
**(b)** Calculate the length of $XY$.

$XY =$ ..... cm **[3]**

**(c)** Calculate the length of the chord $PQ$.

$PQ =$ ..... cm **[4]**

**21** $TA$ and $TB$ are tangents to a circle, centre $O$ and radius $r$, at the points $A$ and $B$.
The line $TO$ intersects the chord $AB$ at $M$.
Angle $ATO$ = Angle $BTO = x$.




**(a)** Show that the length of the chord $AB$ is $2r\cos x$.

**[2]**

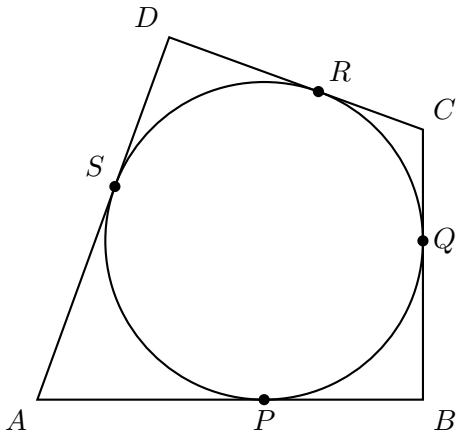
**(b)** A second chord $BC$ is drawn such that $BC = AB$.
Write down the shortest distance from $O$ to the chord $BC$, in terms of $r$ and $x$.

..... **[1]**

**(c)** The area of triangle $OAB$ is equal to the area of triangle $ATB$.
Find the exact value of $x$.

$x =$ ..... $^\circ$ **[5]**

**22** A circle is inscribed in a quadrilateral $ABCD$.
The circle touches the sides $AB$, $BC$, $CD$, and $DA$ at the points $P$, $Q$, $R$, and $S$ respectively.




**(a)** Write down the geometrical reason why $AP = AS$.

..... **[1]**

**(b)** Show that $AB + CD = AD + BC$.

**[3]**

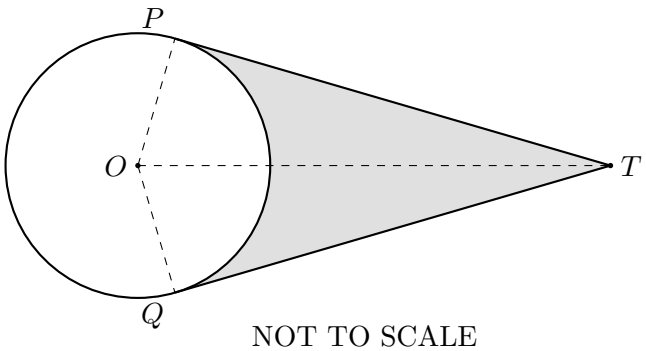
**(c)** Given that $AB = (2x + 5)$ cm, $BC = (3x - 2)$ cm, $CD = (x + 8)$ cm and $AD = (2x + 1)$ cm, work out the value of $x$.

$x =$ ..... **[3]**

**(d)** Calculate the perimeter of the quadrilateral $ABCD$.

Perimeter = ..... $cm$ **[2]**

**23** The diagram shows a circle with centre $O$ and radius 7 cm.
The lines $TP$ and $TQ$ are tangents to the circle at the points $P$ and $Q$.
The distance from the centre $O$ to the external point $T$ is 25 cm.




NOT TO SCALE

**(a)** Show that the length of the tangent $TP$ is 24 cm.

**[2]**

**(b)** Calculate angle $PTQ$.

Angle $PTQ$ = ..... ° **[3]**

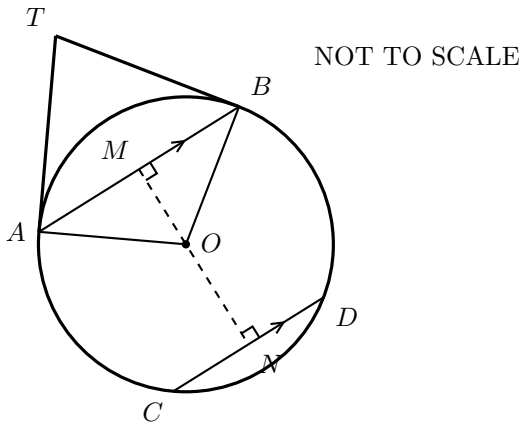
**(c)** Calculate the area of the shaded region bounded by the tangents $TP$, $TQ$ and the minor arc $PQ$.

Area =

.....

$\text{cm}^2$ **[4]**

**24** The diagram shows a circle, centre $O$, of radius $r$ cm. Chords $AB$ and $CD$ are parallel and lie on opposite sides of the centre $O$.
$AB = 16$ cm, $CD = 12$ cm, and the perpendicular distance between $AB$ and $CD$ is 14 cm. The tangents to the circle at $A$ and $B$ meet at the external point $T$.




**(a)** The perpendicular line through $O$ meets $AB$ at $M$ and $CD$ at $N$, with $OM = x$ cm. Write down two equations in terms of $r$ and $x$, and solve them to show that $x = 6$ and $r = 10$.

**[4]**

**(b)** Calculate the exact length of tangent $TA$, giving your answer as an improper fraction.

$TA =$ ..... cm **[3]**

**(c)** Write down the length of $TB$, and calculate the perimeter of quadrilateral $OATB$, giving your answer correct to 1 decimal place.

$TB =$ ..... , Perimeter = ..... **[2]**

**(d)** Calculate the size of angle $ATB$, giving your answer correct to 1 decimal place.

Angle $ATB =$ ..... $^\circ$ **[2]**

**1** **(a)** Recognises that tangents from an external point are equal in length, so triangle $PAB$ is isosceles: $180 - 44 = 136$ **[M1]**

$136 \div 2 = 68$ **[A1]**

**(b)** Recognises angle between tangent and radius is 90°, setting up sum of angles in quadrilateral $OAPB$: $360 - 90 - 90 - 44$ **[M1]**

136 **[A1]**

**[Back to Question]** *[Total Marks: 4]*

**2** **(a)** Tangents from an external point are equal in length **[B1]**

**(b)** 70 **[B1]**

**(c)** Angle $OAP = 90°$ OR Angle $OBP = 90°$ **[M1]**

140 **[A1]**

**[Back to Question]** *[Total Marks: 4]*

**3** **(a)** Angle $PTO = 56 \div 2 = 28°$ **[M1]**

$180 - 90 - 28 = 62$ **[A1]**

**(b)** Tangents from an external point are equal in length (so $TP = TQ$) **[B1]**

**(c)** 62 **[B1]**

**[Back to Question]** *[Total Marks: 4]*

**4** **(a)** Uses angle sum of quadrilateral $OATB$ with angle $OAT = 90°$ and angle $OBT = 90°$: $360° - 90° - 90° - 44°$ OR $180° - 44°$ **[M1]**

136 **[A1]**

**(b)** Recognises triangle $OAB$ is isosceles ($OA = OB$) and uses $(180° - 136°) \div 2$ **[M1]**

22 **[A1]**

**(c)** Tangents from an external point are equal in length **[B1]**

**[Back to Question]** *[Total Marks: 5]*

**5** **(a)** 90 **[B1]**

The perpendicular from the centre of a circle to a chord bisects the chord (or perpendicular bisector of a chord passes through the centre) **[B1]**

**(b)** Uses midpoint $M$: $AM = \dfrac{16}{2} = 8$ **[B1]**

Applies Pythagoras' theorem: $OM^2 + 8^2 = 10^2$ or $OM = \sqrt{10^2 - 8^2}$ **[M1]**

6 **[A1]**

**[Back to Question]** *[Total Marks: 5]*

**6** **(a)** 24 **[B1]**

**(b)** Identifies $\angle OAP = 90°$ (tangent perpendicular to radius) **[M1]**

$OP^2 = 7^2 + 24^2 = 625 \implies OP = 25$ **[A1]**

**(c)** Let $M$ be the intersection of $OP$ and $AB$. Identifies $\triangle OAM \sim \triangle OPA$ OR uses Area of $\triangle OAP$ OR uses $\cos(\angle AOP)$ **[M1]**

$\cos(\angle AOP) = \frac{7}{25}$ and $OM = 7 \times \cos(\angle AOP)$ **[M1]**

$OM = \frac{49}{25}$ or 1.96 **[A1]**

[Back to Question] *[Total Marks: 6]*

**7** **(a)** $AM = 12$ cm **[B1]**

Reason: the perpendicular from the centre of a circle to a chord bisects the chord **[B1]**

**(b)** $OM^2 = 13^2 - 12^2$ $(= 169 - 144 = 25)$ **[M1]**

5 **[A1]**

**(c)** $ON = 5$ cm **[B1]**

Reason: equal chords are equidistant from the centre of the circle **[B1]**

[Back to Question] *[Total Marks: 6]*

**8** **(a)** $x + 3 = 3x - 5$ **[M1]**

$x = 4$ **[A1]**

**(b)** Perpendicular distance $= 7$ (can be implied) **[B1]**

$\sqrt{25^2 - 7^2}$ OR 24 **[M1]**

48 **[A1]**

[Back to Question] *[Total Marks: 5]*

**9** **(a)** 12 **[B1]**

Tangents from an external point to a circle are equal in length **[B1]**

**(b)** Uses $\angle OAP = 90°$ and Pythagoras' theorem: $OP = \sqrt{5^2 + 12^2}$ **[M1]**

13 **[A1]**

**(c)** Calculates area of two congruent right-angled triangles: $2 \times \left(\frac{1}{2} \times 5 \times 12\right)$ **[M1]**

60 **[A1]**

[Back to Question] *[Total Marks: 6]*

**10** **(a)** $OA = OB$ (radii of the circle) **[B1]**

$PA = PB$ (tangents from an external point are equal in length) **[B1]**

$OP$ is common to both triangles (or angle $OAP$ = angle $OBP = 90°$, tangent to radius $= 90°$), so triangles are congruent (SSS or RHS) **[B1]**

**(b) (i)** Angle $OAP$ = angle $OBP = 90°$ **[M1]**

Angle $AOB = 360° - 90° - 90° - 52° = 128$ **[A1]**

**(ii)** $OA = OB$ so triangle $OAB$ is isosceles **[M1]**

Angle $OAB = \dfrac{180° - 128°}{2} = 26$ **[A1]**

**(c)** 14.3 **[B1]**

[Back to Question] *[Total Marks: 8]*

**11 (a)** Angle $OAP = 90°$ AND Angle $OBP = 90°$ (tangent perpendicular to radius) **[B1]**

Angle $AOB = 360 - 90 - 90 - 46 = 134°$ **[B1]**

**(b)** Triangle $OAB$ is isosceles ($OA = OB$) **[M1]**

Angle $OAB = (180 - 134) \div 2 = 23°$ **[A1]**

[Back to Question] *[Total Marks: 4]*

**12 (a)** Angle $OMB = 90°$ (line from centre to midpoint of chord is perpendicular to chord) **[B1]**

$MB = 7$ **[B1]**

$OM = \sqrt{8.5^2 - 7^2}$ **[M1]**

4.77 or 4.769... **[A1]**

**(b)** Half chord $= \sqrt{8.5^2 - 3^2}$ **[M1]**

$2 \times \sqrt{63.25} = 15.9$ or 15.90... **[A1]**

[Back to Question] *[Total Marks: 5]*

**13 (a)** Identifies right-angled triangle $OAM$ (or $OBM$) and states $AM = 24 \div 2 = 12$ **[B1]**

Applies Pythagoras' theorem: $OM^2 + 12^2 = 13^2$ **[M1]**

$OM = \sqrt{169 - 144} = \sqrt{25} = 5$ **[A1]**

**(b)** 24 (using property that chords equidistant from the centre are equal in length) **[B1]**

**(c)** $\dfrac{1}{2} \times 24 \times 5$ **[M1]**

60 **[A1]**

[Back to Question] *[Total Marks: 6]*

**14 (a)** The line from the centre of a circle to the midpoint of a chord is perpendicular to the chord. **[B1]**

**(b)** $AM = 24 \div 2 = 12$ cm **[B1]**

$OM^2 + 12^2 = 13^2$ **[M1]**

$OM = \sqrt{169 - 144} = 5$ **[A1]**

**(c)** $\sin(\angle AOM) = \frac{12}{13}$ OR $\cos(\angle AOM) = \frac{5}{13}$ OR $\tan(\angle AOM) = \frac{12}{5}$ **[M1]**

$\angle AOM = 67.38...°$ **[A1]**

Angle $AOB = 2 \times 67.38... = 134.8°$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**15** **(a)** States or uses $OM \perp AB$ (perpendicular bisector of chord passes through centre), so $AM = 7$ **[B1]**

Uses Pythagoras' Theorem: $OM = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$ **[B1]**

**(b)** States $ON = OM = 24$ (equal chords are equidistant from centre) **[B1]**

Uses Cosine rule: $MN^2 = 24^2 + 24^2 - 2(24)(24)\cos 100°$ OR uses $MN = 2 \times 24 \times \sin 50°$ **[M1]**

36.8 (accept $36.77\ldots$) **[A1]**

**(c)** $\frac{1}{2} \times 24 \times 24 \times \sin 100°$ **[M1]**

284 (accept $283.6\ldots$) **[A1]**

[Back to Question] *[Total Marks: 7]*

**16** **(a)** Forms a right-angled triangle with hypotenuse 13 and base $\frac{24}{2} = 12$ **[M1]**

Uses Pythagoras' theorem: $\sqrt{13^2 - 12^2}$ **[M1]**

5 **[A1]**

**(b)** 5 **[B1]**

**(c)** Uses Pythagoras' theorem: $\sqrt{13^2 - 12^2}$ **[M1]**

Half of chord $EF = 5$ **[A1]**

10 **[A1]**

**(d)** The perpendicular from the centre to a chord bisects the chord (accept "perpendicular bisector of a chord passes through the centre") **[B1]**

[Back to Question] *[Total Marks: 8]*

**17** **(a)** Equal chords are equidistant from the centre, so $OM = ON$. **[B1]**

$5x - 7 = 2x + 2$ **[M1]**

$3x = 9 \implies x = 3$ **[A1]**

**(b)** $OM = 5(3) - 7 = 8$ cm **[B1]**

Uses Pythagoras' theorem: $AM^2 + 8^2 = 17^2$ **[M1]**

$AM = \sqrt{289 - 64} = \sqrt{225} = 15$ **[A1]**

$AB = 2 \times 15 = 30$ **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**18** **(a)** Angle $OTA = 90°$ (tangent perpendicular to radius) **[B1]**

$AT = 12$ (perpendicular from centre to chord bisects chord) **[B1]**

In triangle $OTA$, $r^2 + 12^2 = R^2$ (Pythagoras theorem) **[M1]**

$R^2 - r^2 = 144$ **[A1]**

**(b)** Area $= \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$ **[M1]**

$k = 144$ **[A1]**

**(c)** Distance $= r$ (or same distance as chord $AB$) **[B1]**

Equal chords are equidistant from the centre **[B1]**

**[Back to Question]** *[Total Marks: 7]*

**19** **(a)** The perpendicular bisector of a chord passes through the centre $O$, so $OP \perp AB$ and $OQ \perp CD$ **[B1]**

Since $AB$ and $CD$ are parallel, both $OP$ and $OQ$ are perpendicular to the same parallel lines through $O$, forming a single straight line $POQ$ **[B1]**

**(b)** Identifies $AP = 12$, $CQ = 5$, and $OQ = 17 - x$ **[B1]**

Sets up Pythagoras' equations for both triangles: $r^2 = x^2 + 12^2$ and $r^2 = (17 - x)^2 + 5^2$ **[M1]**

Equates expressions for $r^2$: $x^2 + 144 = 289 - 34x + x^2 + 25$ **[M1]**

Solves linear equation: $34x = 170$ giving $x = 5$ **[A1]**

Substitutes $x = 5$ to find $r = \sqrt{5^2 + 12^2} = 13$ **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**20** **(a)** The perpendicular bisector of a chord passes through the centre of the circle **[B1]**

**(b)** $MX = 16 \div 2 = 8$ **[B1]**

$OX = \sqrt{10^2 - 8^2} = 6$ **[M1]**

$10 - 6 = 4$ **[A1]**

**(c)** Distance of $PQ$ from $O = 11 - 6 = 5$ **[B1]**

Half length of $PQ = \sqrt{10^2 - 5^2}$ **[M1]**

$\sqrt{75}$ or 8.66... **[A1]**

$2 \times \sqrt{75} = 17.3$ (17.32...) **[A1]**

**[Back to Question]** *[Total Marks: 8]*

**21** **(a)** Identifies $\angle OAT = 90°$ and $\angle AOT = 90° - x$ **[M1]**

Uses $AM = r\sin(90° - x) = r\cos x$ and perpendicular bisector property $AB = 2AM$ to get $2r\cos x$ **[A1]**

**(b)** $r\sin x$ (equal chords are equidistant from centre, distance $= OM = r\cos(90° - x)$) **[B1]**

**(c)** Area of $OAB = \frac{1}{2} \times AB \times OM = \frac{1}{2}(2r\cos x)(r\sin x) = r^2 \sin x \cos x$ **[M1]**

$TM = \frac{AM}{\tan x} = \frac{r\cos x}{\tan x} = \frac{r\cos^2 x}{\sin x}$ **[M1]**

Area of $ATB = \frac{1}{2} \times AB \times TM = \frac{1}{2}(2r\cos x)\left(\frac{r\cos^2 x}{\sin x}\right) = \frac{r^2\cos^3 x}{\sin x}$ **[M1]**

Equates areas: $r^2 \sin x \cos x = \frac{r^2\cos^3 x}{\sin x} \implies \sin^2 x \cos x = \cos^3 x \implies \tan^2 x = 1$ **[M1]**

45 **[A1]**

[Back to Question] *[Total Marks: 8]*

**22** **(a)** Tangents from an external point are equal in length **[B1]**

**(b)** Recognises $BP = BQ$, $CR = CQ$, and $DR = DS$ **[M1]**

$AB + CD = AP + PB + CR + RD$ **[M1]**

Substitutes equal tangent segments:
$AS + BQ + CQ + DS = (AS + DS) + (BQ + CQ) = AD + BC$ **[A1]**

**(c)** Substitutes expressions into the relationship from part (b):
$(2x + 5) + (x + 8) = (2x + 1) + (3x - 2)$ **[M1]**

Correctly expands and simplifies to a linear equation, e.g., $3x + 13 = 5x - 1$ **[M1]**

7 **[A1]**

**(d)** Substitutes $x = 7$ into the lengths of the sides (at least two) OR uses $2(3x + 13)$ **[M1]**

68 **[A1]**

[Back to Question] *[Total Marks: 9]*

**23** **(a)** Recognises angle $OPT = 90°$ and sets up Pythagoras: $TP^2 + 7^2 = 25^2$ **[M1]**

$TP = \sqrt{625 - 49} = \sqrt{576} = 24$ **[A1]**

**(b)** Uses trigonometry in triangle $OPT$: $\sin(\text{angle } PTO) = \frac{7}{25}$ or $\tan(\text{angle } PTO) = \frac{7}{24}$ **[M1]**

Angle $PTO = 16.26°$ **[A1]**

Angle $PTQ = 2 \times 16.26° = 32.5°$ (accept $32.52°$) **[A1]**

**(c)** Calculates total area of kite $OPTQ = 2 \times (\frac{1}{2} \times 7 \times 24) = 168$ **[M1]**

Calculates angle $POQ = 360 - 90 - 90 - 32.52 = 147.48°$ (or uses $180 - 32.52$) **[M1]**

Calculates area of sector $OPQ = \frac{147.48}{360} \times \pi \times 7^2 = 63.07\ldots$ **[M1]**

Shaded area $= 168 - 63.07 = 104.93$ (accept 104.9 to 105.0) **[A1]**

[Back to Question] *[Total Marks: 9]*

**24** **(a)** $AM = 8$ cm and $CN = 6$ cm (perpendicular from centre bisects chord), giving $r^2 = x^2 + 8^2$ **[B1]**

$r^2 = (14 - x)^2 + 6^2$ **[B1]**

$x^2 + 64 = 196 - 28x + x^2 + 36$ leading to $28x = 168$ and $x = 6$ **[M1]**

$r = \sqrt{6^2 + 8^2} = 10$ **[A1]**

**(b)** Angle $OAT = 90°$ (tangent to radius $= 90°$) and $\tan(\text{angle } AOM) = \dfrac{8}{6}$ **[M1]**

$TA = 10 \times \dfrac{8}{6}$ (or similar triangles $OMA$ and $OAT$ used correctly) **[M1]**

$\dfrac{40}{3}$ **[A1]**

**(c)** $TB = \dfrac{40}{3}$ cm (or 13.3 cm) **[B1]**

Perimeter $= 10 + 10 + \dfrac{40}{3} + \dfrac{40}{3} = 46.7$ cm **[B1]**

**(d)** $\tan(\text{angle } ATO) = \dfrac{10}{40/3} = \dfrac{3}{4}$ giving angle $ATO = 36.87°$ **[M1]**

Angle $ATB = 2 \times 36.87° = 73.7$ **[A1]**

**[Back to Question]** *[Total Marks: 11]*